

Nicholas Ketz, PhD

Applied Research Scientist: Artificial Intelligence, Continual Learning, Bioinformatics, Interpretability

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Summary

I am an applied scientist rooted in computational cognitive neuroscience. I have developed novel solutions to complex problems across multiple domains for nearly 20 years, with roles spanning the full AI lifecycle in both academic and industrial research labs, and a continuing focus on understanding what learned systems have internalized and constraining how they act.

Relevant Experience

Concurrent research threads across roles listed in Education/Employment below.

Large Language Modeling

03/2022 - Present

- Developing and deploying DNA Large Language Models (Evo, ESM, Enformer/Borzo-class regulatory models) for cell and gene therapy optimization, cross-species engineering, and viral vector/transgene packaging design

Generative AI

01/2020 - 01/2022

- Created agent-based generative AI systems for video/image generation and autonomous-driving domain adaptation, grounded in learned human concepts (patented, US 12,106,779)

Autonomous System Safety & Evaluation

09/2018 - 01/2022

- Patented a safety-and-efficacy override system for autonomous systems, using competency assessment and prospective simulation to trigger handover to a human or assistive system when predicted competency falls below a safety threshold (US 11,420,655)
- Co-developed a domain-agnostic metrics framework for evaluating continual learning systems under distribution shift, applied to five independently-developed lifelong learning systems across 15 collaborating institutions (DARPA L2M program)
- Developed A-EMS, an online method for autonomous agents to safely respond to out-of-distribution situations in real time (TAROS 2022)

Lifelong Deep Learning

09/2018 - 01/2021

- Key personnel in DARPA's L2M program, developed methods to preserve performance across large sequential distribution shifts
- Co-developed a world-model-based pseudo-rehearsal method for continual learning, later forming the basis of the Episodic World Model used in the safety-override system above (arXiv, 2019)

Closed-loop Neural Stimulation

09/2016 - 06/2018

- Developed devices and algorithms for closed-loop neural stimulation in human subjects to enhance cognitive performance
- Closed-loop transcranial stimulation system enhanced sleep-dependent memory generalization by modulating endogenous slow-wave oscillations (Journal of Neuroscience, 2018; DARPA RAM-Replay program)

Education/Employment

Principal AI/ML Scientist *Colossal Biosciences/Form Bio*

02/2022 - Present

- Build LLM-agent pipelines that query SRA, GEO, and ENCODE, assemble evidence, and score candidate designs against versioned rubrics
- Advise on technical direction for a cross-disciplinary science team and lead development directly on high-stakes projects

Research Scientist: Information and Systems Sciences *HRL Laboratories*

09/2016 - 01/2022

- Developed novel AI methods funded by Department of Defense with product development for aerospace and automotive industries
- Contributed to DARPA's RAM-Replay program and internally-funded research exploring model-based reinforcement learning

PhD: Computational Cognitive Neuroscience *University of Colorado, Boulder*

09/2010 - 09/2016

- Developed biologically inspired neural networks informed by neuroimaging experiments of cognitive function, advised by Randall O'Reilly
- Co-authored a review proposing a thalamic model of prefrontal cortex–medial temporal lobe interactions in long-term memory (Trends in Neurosciences, 2015)

Research Assistant: Davachi Human Memory Lab *New York University*

09/2007 - 07/2010

- Supported NIH R01 research for neuroimaging of the human long-term memory system
- Contributed to fMRI research on hippocampal-cortical connectivity during rest, published in Neuron (2010)

BA: Physics, minor: Psychology *University of Minnesota, Twin Cities*

09/2003 - 06/2007

- Studied physics with a minor in psychology, with research experience in particle physics and cognitive neuroscience

Skills

- **Technical:** Deep Learning (CNN, RNN, Transformers, RL), Machine Learning (Bayesian Optimization, Gaussian Processes, Unsupervised Learning), Statistics (Bayesian, Parametric/Non-parametric, Time-series), Programming (Python, PyTorch, TensorFlow, scikit-learn, MATLAB, R, Docker, Kubernetes, Git, Bash, SQL), Agentic AI (LLM agents, tool-use orchestration, agentic coding workflows), Cloud (AWS, GCP)
- **Interpretability:** Integrated Gradients, Captum, TF-MoDISco motif discovery, validating learned representations against known domain mechanisms
- **NeuroImaging:** fMRI (event-related, resting-state, connectivity), EEG (event-related, sleep, source localization), transcranial electrical stimulation (tACS/tDCS), experimental design for human studies
- **AI Lifecycle:** Full-cycle AI development (proof-of-concept to production), cross-team product development, agentic workflows for code and product delivery, containerized system design, CI/CD pipelines, cross-container unit testing
- **Professional:** Team leadership, cross-departmental collaboration (engineering, product, sales), client/stakeholder management, project coordination across multiple research domains and organizations

Select Publications/Patents/White Papers

System and method for safety and efficacy override of an autonomous system *US Patent, 2022*

- Bounding autonomous behavior via predicted competency using a learned world model, applied to autonomous driving systems.

A-EMS: Adaptive Emergency Management for Autonomous Agents *TAROS, 2022*

- Related sample-efficient extension of the efficacy override framework to real-time out-of-distribution response using Bayesian optimization.

A domain-agnostic approach for characterization of lifelong learning systems *Neural Networks, 2023*

- Metrics suite adopted across the DARPA L2M program's five independently-developed systems

Leveraging Model Interpretability Methods to Predict Gene Therapy Manufacturing Failures *White Paper, 2023*

- Sole-authored; used Integrated Gradients (Captum) and TF-MoDISco to show a CNN's truncation predictions are related to non-canonical DNA secondary structures, independently validating the model's learned representation

Sliced Cramér synaptic consolidation for preserving deeply learned representations *ICLR, 2020 (Spotlight)*

- Preserves a network's learned output distribution at an arbitrary layer to prevent catastrophic forgetting